Nemo-NAA10km : A NEMO based ocean modelling configuration for North Atlantic & Arctic

June 18, 2026

1 Overview

Nemo-NAA10km is a NEMO based regional ocean modelling configuration. Its domain covers the North Atlantic ocean from a latitude of 35°N, the Nordic Seas, the Arctic Ocean & a piece of Pacific Ocean. Its resolution is 10km, with increased resolution up to 8km in the Arctic (Figure 1). The purpose of this note is to present its features and results. However, an updated version is being produced as these lines are being written, which should resolve some bias especially related with a too high salinity of the Atlantic Water flow towards the Arctic.

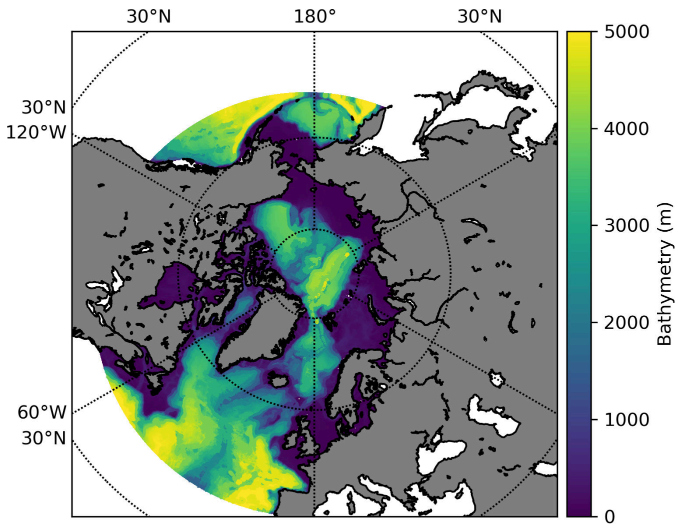

Figure 1: Domain and bathymetry (in m) of the Nemo-NAA10km configuration

Its interaction with the atmosphere works in forced mode only, and it has two open boundary conditions (one in the Northern Atlantic and one in the Northern Pacific). Nemo-NAA10km is fully two-way coupled with the SI3 sea-ice model. Nemo-NAA10km is not a climate model, but an ocean process model aimed at studying the underlying physics of processes, and process changes. For this purpose Nemo-NAA10km is designed to provide realistic results without any form of data assimilation or restoring. Its only form of control on temperature or salinity is made at its open boundary conditions. And it is through a thorough representation of each physical process, that it achieves a realistic representation of temperature and salinity fields within its domain, even if long term simulations of more than 50 years are considered. Nemo-NAA10km works in hindcast mode, forced by re-analysis (ERA5 for the atmosphere and GLORYS for the ocean for example), or in climate downscaling mode forced by climate model outputs (NorESM or EC-Earth for example). A summary of Nemo-NAA10km features can be found in Appendix I.

The design of Nemo-NAA10km started in 2018, and used the Nemo-Nordic setup as inspiration (Link to Nemo-Nordic reference paper). Nemo-NAA10km was first based on NEMO v3.6, and is now based on NEMO v4.2.2. Switching to new versions of NEMO permits to profit from the most recent scientific and technical developments of the NEMO ocean engine. Nemo-NAA10km is refered to in more than 12 peer-reviewed articles in various journals (JGR-Oceans, Nature CC, GMD etc...). Recent developments made on Nemo-NAA10km include the online coupling with the biogeochemical model PISCES, and the use of the AGRIF tool to increase locally the resolution up to 1km. These two developments were demonstrated to work together. The analysis of these simulation is ongoing.

The NEMO ocean engine is a ocean modelling toolbox used, develloped and maintained by a huge community of researchers, engineers and technicians worldwide (Link to the NEMO ocean engine website).

2 Evaluation Procedure

The evaluation procedure of Nemo-NAA10km is based on the long experience of physical oceanography of IMR researchers, associating both experts in observations and ocean modelling. The evaluation procedure is based on the results of a simulation forced by ERA5 from 1942 to present, which uses the GLORYS re-analysis at the open boundary conditions of the domain. The part of the simulation used for evaluation is the period 1993-2024. It is important to note that when the evaluation is done, the model has been running without any safety net for more than 50 years: its only form of control on the hydrography is done at its boundary conditions.

The first validation step is sea level (SSH), for which a good representation is the sine qua non condition for an ocean model to be realistic. This validation is made against tide gauges at high frequency to ensure the representation of barotropic waves, which carry energy in the model, at different velocities. Validation is made against tide gauges along the Norwegian coast, using hourly output over several years. Correlation and standard deviation are computed with and without tidal signals included. The correlation of SSH including tides reaches more than 95%, and 80% without tides. Baroclinic validation, sea ice cover and trends are described based on several campaign of observations, and are still updated as these lines are written. A first evaluation was provided in Nemo-NAA10km first reference article (Link to Nemo-NAA10km reference paper). The ongoing evaluation carried by IMR, and takes the form of a long document from which only a few extracts are provided in the present sketch.

2.1 Comparison with tide gauges

We provide hereafter a comparison of wind driven hourly SSH for the period 1993-2024, for three tide gauges located along the Norwegian coast. The comparison of standard deviations between observations and model outputs shows that there is a margin of improvement, Nemo-NAA10km's standard deviation is always too high in comparison with observations, which indicates that the mean bottom roughness is too low.

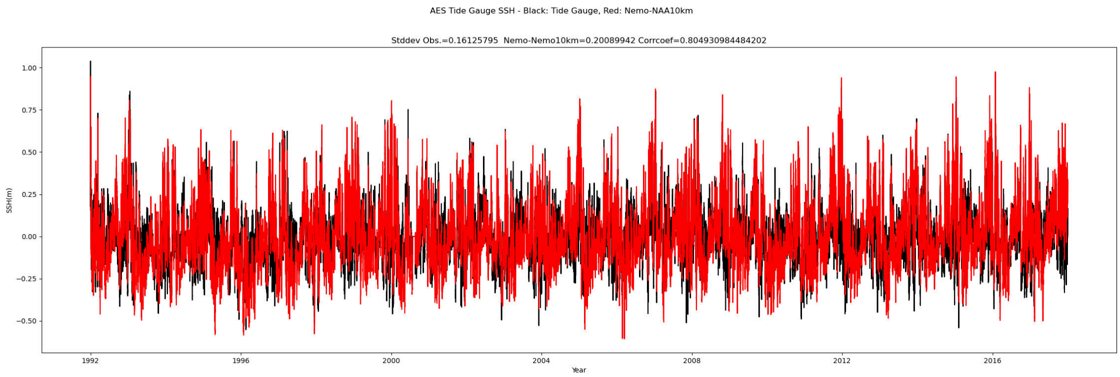

Figure 2: Ålesund - SSH comparison with hourly tide gauges, only with wind driven SSH.

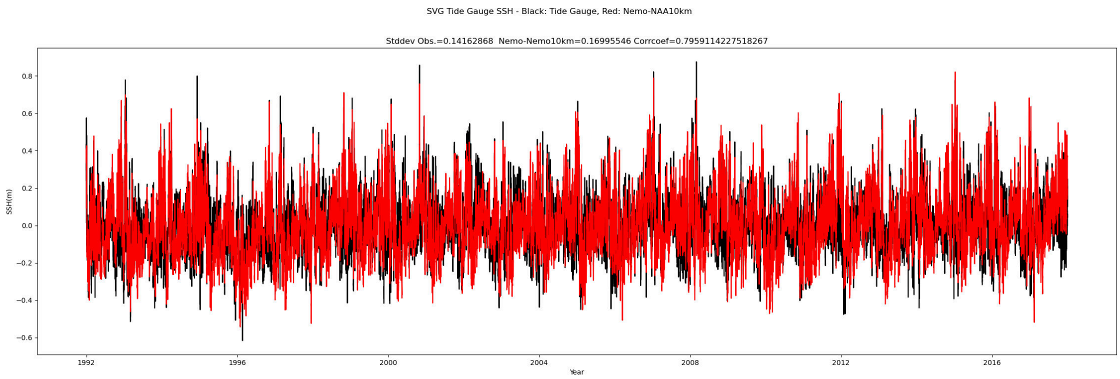

Figure 3: Stavanger - SSH comparison with hourly tide gauges, only with wind driven SSH.

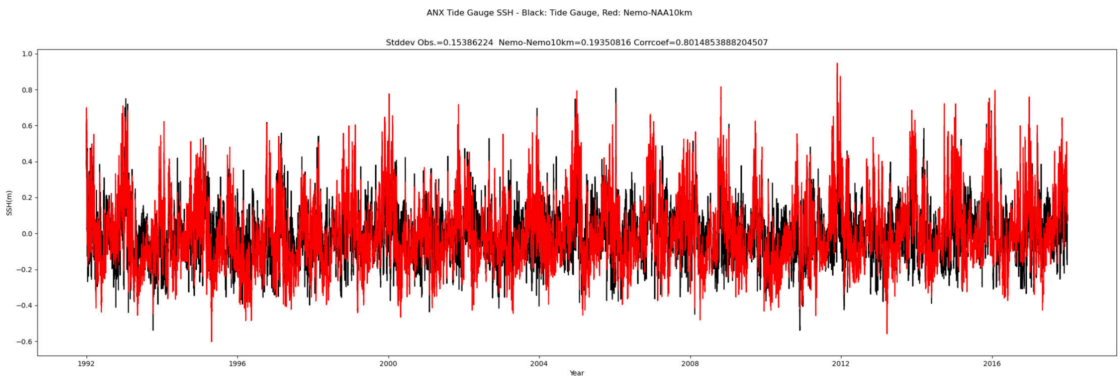

Figure 4: Andenes (Lofoten) - SSH comparison with hourly tide gauges, only with wind driven SSH.

2.2 Section Comparison

2.2.1 Gimsøy cross section comparison

The Gimsøy section extends perpendicularly from the Norwegian coast at the latitude of the Lofoten islands. It captures three different water masses that are critical to describe the overall behaviour of the model. The Atlantic Water (AW hereafter), that is warmer and saltier, is also lighter than the surrounding Norwegian Sea water. The AW progression towards the Arctic is possible only if it stays at the surface, while losing its heat content. At the surface of the Nordic Seas, a layer of fresher water can be found, which originates from the East Greenland Current (Westwards of the AW flow), or the Norwegian Coastal Current (Eastwards of the AW flow). The surrounding Nordic Sea water is colder and fresher than the AW flow, is exhibits colder water originating in the Arctic at the bottom of the basin.
The comparison at Gimsøy is done with observational campaign led by IMR, but also with outputs of the Norkyst ROMS based ocean model, which has open boundary conditions in the Arctic Northward of the Nordic Seas, and Southward of the Greenland-Scotland ridge, hence giving a stronger control over water masses.

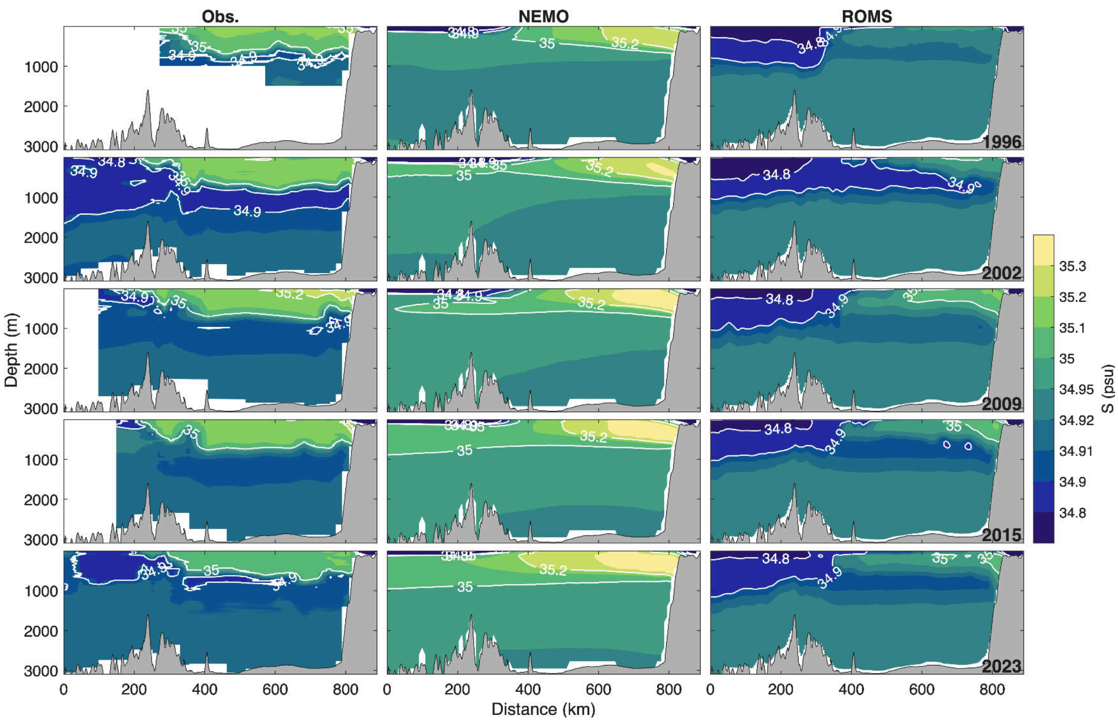

Figure 5: Comparison of salinity between observations, Nemo-NAA10km and the ROMS SVIM configuration

Nemo-NAA10km reproduces the different water masses. The AW mass, for which the only form of external control is the open boundary condition located very far away from the section, appears usually with a slight positive salinity/temperature bias. The East Greenland Current freshwater water has the right extent and low salinity, but appears too stratified, and does not mix deep enough with the Nordic Seas water mass. Finally, the salinity and temperature at the

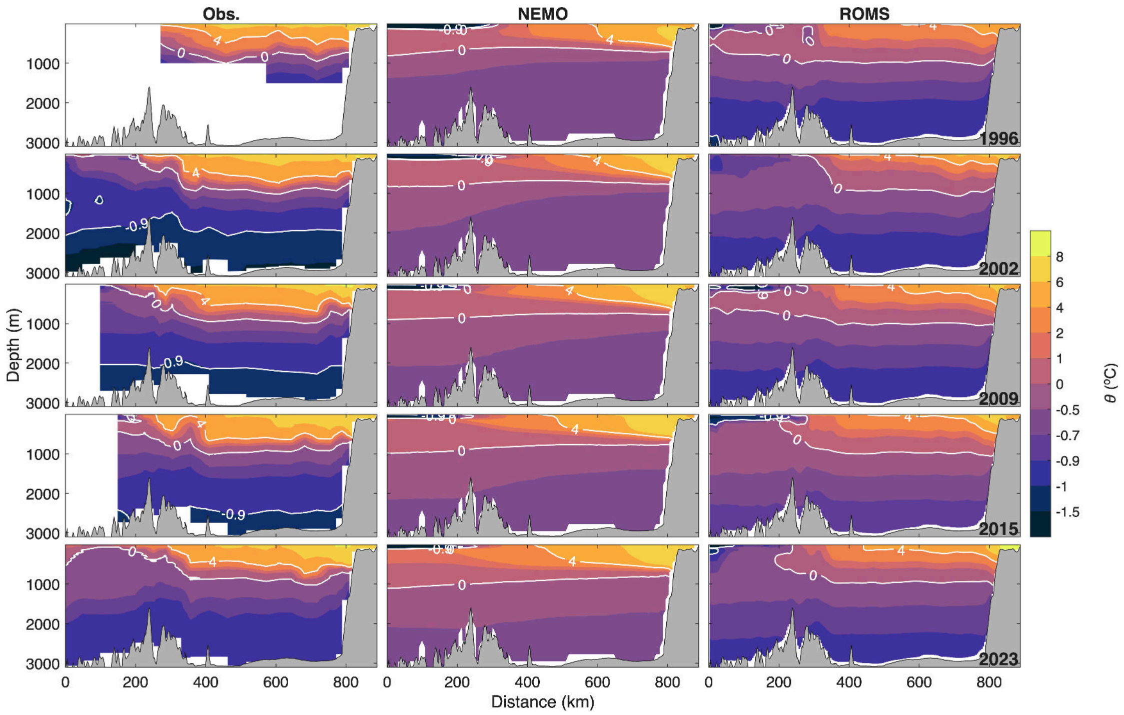

Figure 6: Comparison of temperature between observations, Nemo-NAA10km and the ROMS SVIM configuration

bottom of the section are slightly over estimated.

2.2.2 Arctic cross section comparison

We also compare Nemo-NAA10km with several cross sections located in the Arctic Ocean. As before Nemo-NAA10km is compared with the observations, and with the ROMS based SWIM ocean model.

At Barents Sea Opening, one notices also a sligh positive salinity bias, as for the Gimsøy section (Figures 7). This Atlantic Water positive bias propagates into the Barents Sea (Figure 8) but becomes less pronounced as the Atlantic Water progresses into the Arctic. The shelf break located North of the Svalbard archipelago shows that the core of Atlantic Water propagation through the Fram Strait branch is also represented (Figure 9).

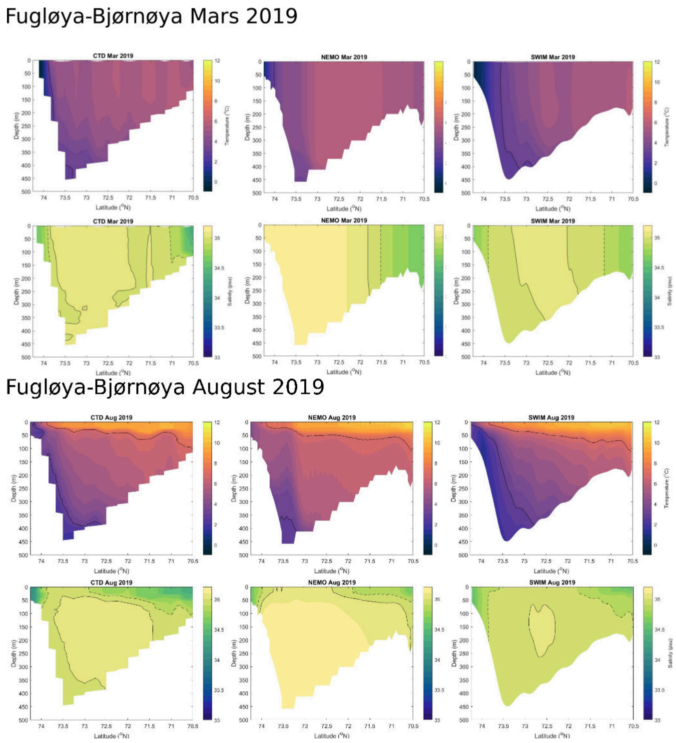

Figure 7: Barents Sea Opening, March and August 2019. Comparison of temperature between observations, Nemo-NAA10km and the ROMS SVIM configuration.

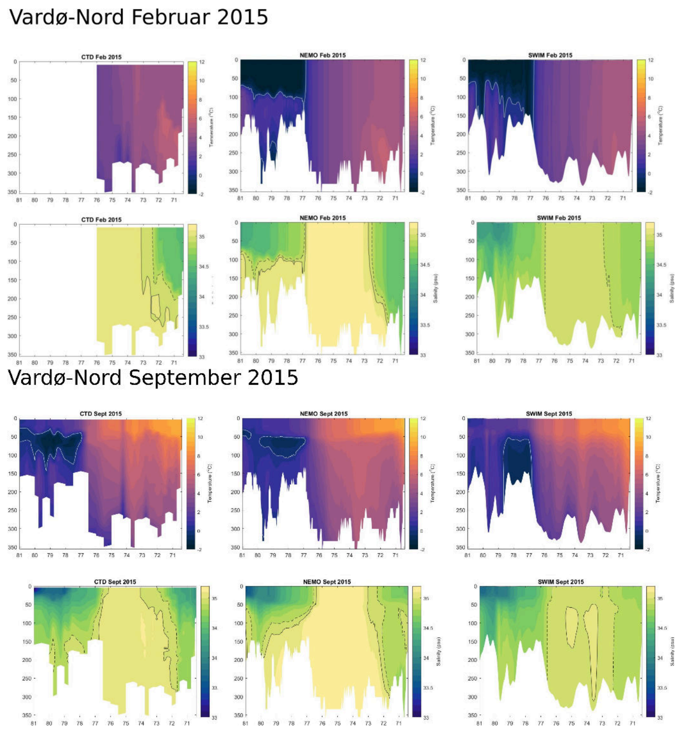

Figure 8: Vardø North Section (Barents Sea). Comparison of Nemo-NAA10km with observations and with the ROMS SVIM configuration. Months of February and September 2015.

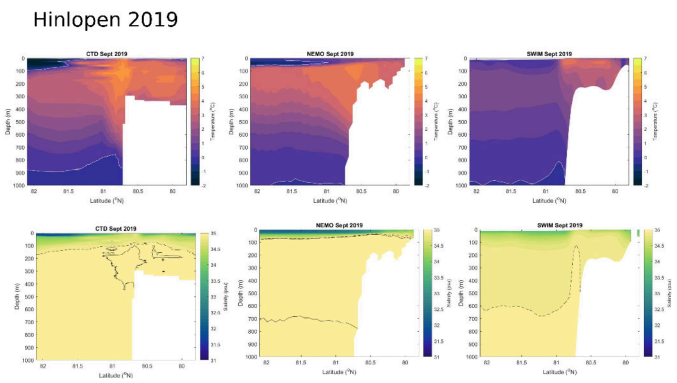

Figure 9: Hinlopen Section (North Svalbard). Comparison of Nemo-NAA10km with observations and with the ROMS SVIM configuration. Months of September 2019.

2.3 Comparison with the GLORYS re-analysis

Mean surface values and budgets from surface to 700m are made for different areas of the North Atlantic & Arctic (Figure 10). The comparison also includes the TOPAZ model which has a similar geographical domain, but for which 1993 is the starting year. The TOPAZ simulation presented here is on its way to be upadted as these lines are being written. The GLORYS re-analysis is always the red line, Nemo-NAA10km is in light green, and TOPAZ is in blue. The results presented here are only time series, but other results such as trend maps or seasonal cycles are available. Nemo-NAA10km produces a good representation of water mass characteristics, although a positive salinity bias is noticed in the Southern part of the computation domain.

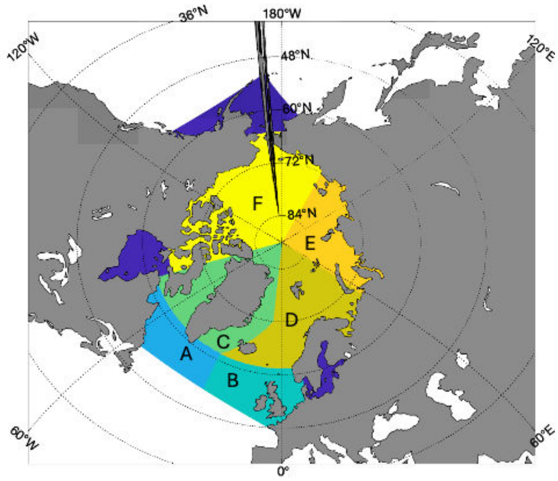

Figure 10: Integration sub-domains designed for diagnostics

2.3.1 Surface Values

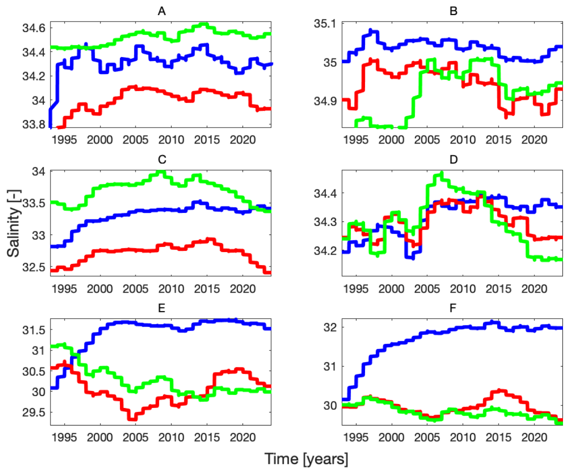

Figure 11: Mean Surface Salinity values for the integration sub-domains (Figure 10)

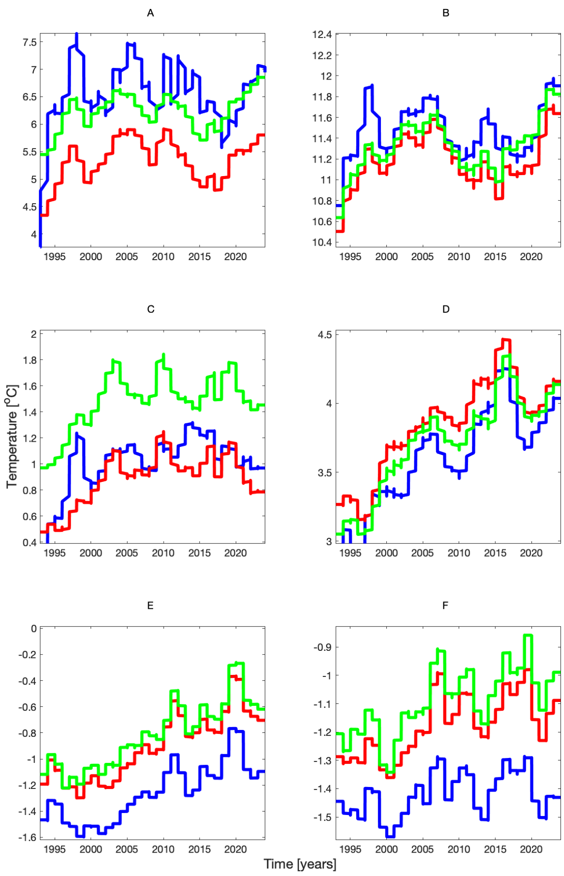

Figure 12: Mean Surface Salinity values for the integration sub-domains (Figure 10)

2.3.2 Water Mass Values

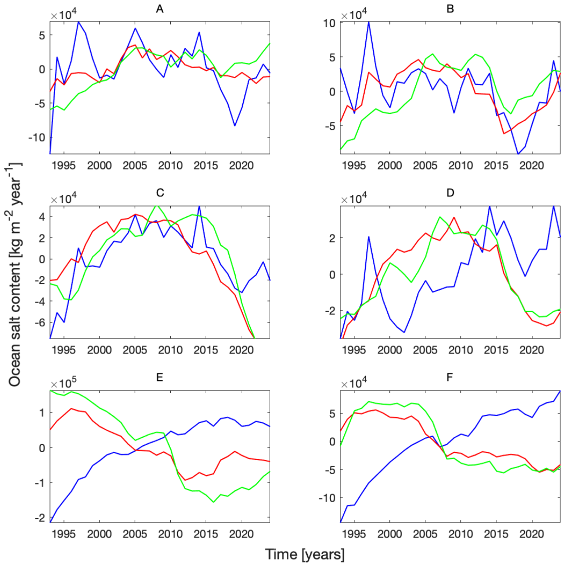

Figure 13: Integrated 0-700m depth salt content values for the integration sub-domains (Figure 10)

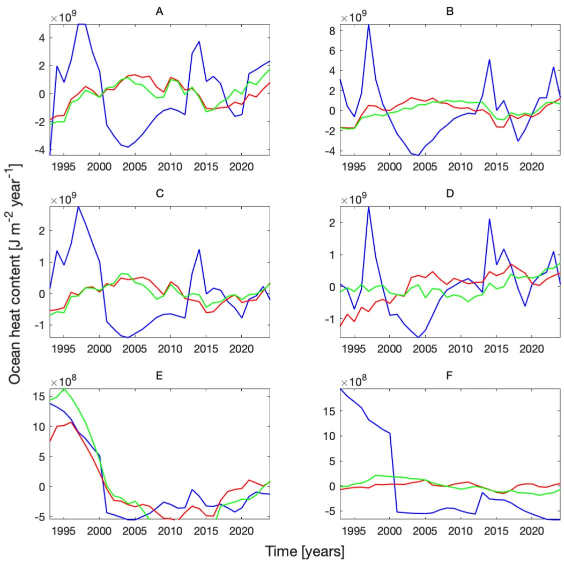

Figure 14: Integrated 0-700m depth heat content values for the integration sub-domains (Figure 10)

Appendix: Summary of Nemo-NAA10km features in hindcast mode

Table 1: Main features of Nemo-NAA10km

Domain	North Atlantic down to a latitutde of 35N (strait zonal open boundary condition between Southern Spain and US), Arctic Ocean and a piece of Pacific Ocean (zonal boundary at 50N).
Description	Nemo-NAA10km is based on NEMO 4.2.2, it uses the SI3 sea-ice model. Plans are being made to switch to Nemo 5.x depending on updates.
Vertical grid	Nemo-NAA10km uses 75 levels of z^* coordinates (SSH time varying z coordinates), and a non-linear free surface.
Coupling	Full two way coupling between ocean and sea-ice. The model interaction with atmospheric data is made through bulk formulation.
Initial condition	The initial condition is a rest state with climatological values of temperature and salinity for year 1942. The period 1942-1950 is considered as a spin up time.
Open boundary conditions	GLORYS based open boundary conditions for temperature, salinity and SSH.
Atmospheric forcing	ERA5 from 1942, bulk formulation for fluxes.
River runoff	Dai and Trenberth climatology, with modulations
Tidal forcing	TPXO SSH and current data at open boundary conditions + tidal potential over the domain.
Solar radiation	Provided by ERA5
Computational ressources	Betzy computer (Sigma2) and Fahrenheit computer at NSC Linköping.

List of peer reviewed publications using Nemo-NAA10km (Ref. 11 can be considered as a validation reference) :

1. Revisiting the parameterization of dense water plume dynamics in geopotential coordinates in NEMO v4.2.2 *Geosci. Model Dev., 19, 2677–2690, https://doi.org/10.5194/gmd-19-2677-2026, 2026. Robinson Hordoir, Jarle Berntsen, Magnus Hieronymus, Per Pemberton, and Hjalmar Hatun*

2. Barents Sea atlantification driven by a shift in atmospheric synoptic timescale, 2026, *Nature Climate Change volume 16, pages 179–186, Robinson Hordoir, Vahidreza Jahanmard, Pål Erik Isachsen, Ulrike Löptien, Heiner Dietze, Anne Britt Sandø & Vidar S. Lien*

3. Future poleward distribution shifts of community and functional groups in the Barents Sea modelled under different climate and fisheries scenarios, 2025, *Marine Ecology Progress Series, DOI: 10.3354/meps14868, Marcela C. Nascimento, Filippa Fransner,* Robinson Hordoir*, Morten D. Skogen, Raul Primicerio, Torstein Pedersen*

4. Barotropic Trends Through the Barents Sea Opening for the Period 1975–2021, 2025, *Journal of Geophysical Research: Oceans 130(1), DOI: 10.1029/2024JC021663, Vahidreza Jahanmard, Ulrike Löptien, Anne Britt Sandø, Andrea M.U. Gierisch, Heiner Dietze, Vidar S. Lien, Nicole Delpeche-Ellmann, and Robinson Hordoir*

5. A multi-scenario analysis of climate impacts on plankton and fish stocks in northern seas, 2024, *Fish and Fisheries 25(4), DOI: 10.1111/faf.12834, Anne Britt Sandø, Solfrid S. Hjøllo, Cecilie Hansen, Morten D. Skogen, Robinson Hordoir, and Svein Sundby*

6. *Koseki, M., Sandø, A. B., Ottersen, G., Årthun, M., and Stiansen, J. E. 2025. Exploration of short-term predictions and long-term projections of barents sea cod biomass using statistical methods on data from dynamical models. PLOS ONE, 20: 1–28. URL https://doi.org/10.1371/journal.pone.0328762.*

7. *Ma, S., Huse, G., Ono, K., Nash, R. D. M., Sandø, A. B., Nedreaas, K., Sætre Hjøllo, S., et al.Recruitment regime shifts and nonstationarity are widespread phenomena in harvestable stocks experiencing pronounced climate fluctuations. Fish and Fisheries, n/a. URL https://onlinelibrary.wiley.com/doi/abs/10.1111/faf.12810.*

8. Trivial gain of downscaling in future projections of higher trophic levels in the Nordic and Barents Seas, 2023, *Fisheries Oceanography 32(5), DOI: 10.1111/fog.12641, Ina Nilsen, Filippa Fransner, Are Olsen, Jerry Tjiputra, Robinson Hordoir and Cecilie Hansen*

9. dos Santos Schmidt, T. C., Slotte, A., Olafsdottir, A. H., Nøttestad, L., Jansen, T., Jacobsen, J. A., Bjarnason, S., et al. 2023. Poleward spawning of Atlantic mackerel (Scomber scombrus) is facilitated by ocean warming but triggered by energetic constraints. ICES Journal of Marine Science, p. fsad098. URL https://doi.org/10. *1093/icesjms/fsad098.*

10. *Falconer, L., Ytteborg, E., Goris, N., Lauvset, S. K., Sandø, A. B., and Hjøllo, S. S. 2023. Context matters when using climate model projections for aquaculture. Frontiers in Marine Science, 10. URL https://www.frontiersin.org/articles/10.3389/*

fmars.2023.1198451.

11. Changes in Arctic Stratification and Mixed Layer Depth Cycle: A Modeling Analysis, 2022, *Journal of Geophysical Research: Oceans 127(1), DOI: 10.1029/2021JC017270, Robinson Hordoir, Øystein Skagseth, Randi B. Ingvaldsen, Anne Britt Sandø, Ulrike Löptien, Heiner Dietze, Andrea M. U. Gierisch, Karen M. Assmann, Øyvind Lundesgaard and Sigrid Lind*

12. Pan-Arctic suitable habitat model for Greenland halibut, 2021, ICES Journal of Marine Science, 2021, *78(4), DOI: 10.1093/icesjms/fsab007, Mikko Vihtakari, Robinson Hordoir, Margaret Treble, Meaghan D. Bryan, Bjarki Elvarsson, Adriana Nogueira, Elvar H. Hallfredsson, Jørgen Schou Christiansen, and Ole Thomas Albert*